Impact of Blood Flow Restriction Exercise on Muscle Fatigue Development and Recovery

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ABSTRACT

HUSMANN, F., T. MITTLMEIER, S. BRUHN, V. ZSCHORLICH, and M. BEHRENS. Impact of Blood Flow Restriction Exercise on Muscle Fatigue Development and Recovery. *Med. Sci. Sports Exerc.*, Vol. 50, No. 3, pp. 436–446, 2018. **Purpose**: The present study was designed to provide mechanistic insight into the time course and etiology of muscle fatigue development and recovery during and after low-intensity exercise when it is combined with blood flow restriction (BFR). **Methods**: Seventeen resistance-trained males completed four sets of low-intensity isotonic resistance exercise under two experimental conditions: knee extension exercise (i) with BFR and (ii) without BFR (CON). Neuromuscular tests were performed before, during (immediately after each set of knee extension exercise), and 1, 2, 4, and 8 min after each experimental condition. Maximal voluntary torque, quadriceps twitch torque in response to paired electrical stimuli at 10 Hz (PS10) and 100 Hz (PS100), $PS10 \cdot PS100^{-1}$ ratio as an index of low-frequency fatigue, and voluntary activation were measured under isometric conditions. Perceptual and EMG data were recorded during each exercise condition. **Results**: After the first set of exercise, BFR induced significantly greater reductions in maximal voluntary torque, PS100, and $PS10 \cdot PS100^{-1}$ ratio compared with CON. These parameters progressively declined throughout the BFR protocol but recovered substantially within 2 min postexercise when blood flow was restored. Neither a progressive decline in the course of the exercise protocol nor a substantial recovery of these parameters occurred during and after CON. Only at exercise termination, voluntary activation differed significantly between BFR and CON with greater reductions during BFR. **Conclusion**: At the early stage of exercise, BFR exacerbated the development of muscle fatigue mainly due to a pronounced impairment in contractile function. Despite the high level of muscle fatigue during BFR exercise, the effect of BFR on muscle fatigue was diminished after 2 min of reperfusion, suggesting that BFR has a strong but short-lasting effect on neuromuscular function. **Key Words:** CENTRAL FATIGUE, HYPOXIA, METABOLIC STRESS, PERIPHERAL FATIGUE, QUADRICEPS MUSCLE, VASCULAR OCCLUSION

It has been traditionally suggested that muscle growth can only be achieved with high-intensity resistance exercise (70%–80% one-repetition maximum [1RM]), whereas no significant hypertrophic effects were expected after low-intensity exercise. Mechanical stress was therefore considered as the essential stimulus for muscle hypertrophy (1). However, when low-intensity resistance exercise (20%–30% 1RM) was combined with blood flow restriction (BFR), similar gains in muscle mass were observed (2). BFR is used to limit intramuscular oxygen delivery and to prevent venous clearance of metabolites that, in turn, lead to increased metabolic stress (i.e., depletion of phosphocreatine, increased accumulation of inorganic phosphate, protons, and lactate). Even without heavy loads, metabolic stress is thought to trigger certain mechanisms (i.e., systemic hormone production, increased fast-twitch fiber recruitment, and cell swelling), which potentially mediate muscle growth (for a review, see [3]).

Besides the effect of metabolic stress as an obvious key stimulator for hypertrophic adaptations, disturbances in the intra- and extracellular environment induced by BFR exercise are strongly associated with a reduction in maximal voluntary force production (i.e., [4,5]). The impaired force or power-generating capacity of a muscle or muscle group is indicative of muscle fatigue, which stems from a decrease in neural activation of the muscle (i.e., commonly termed "central fatigue") and/or alterations at or distal to the neuromuscular junction that result in contractile dysfunction (commonly termed "peripheral fatigue") (6). Muscle fatigue

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is typically considered to be task dependent (i.e., exercise intensity, duration, contraction mode, active muscle mass) and particularly affected by local and systemic hypoxia (7,8). As might be expected, initial findings by Karabulut et al. (9) have demonstrated that BFR exacerbates the end-exercise level of muscle fatigue after work-matched low-intensity exercise. In detail, higher contributions of peripheral and central factors were responsible for the pronounced muscle fatigue as indicated by reductions in voluntary activation (−13%) and contractile twitch torque (−44%) after BFR exercise compared with the free blood flow condition (+4% and −19%, respectively).

Previous work investigating the effect of BFR on neuromuscular function is currently limited to pre- and post-exercise measurements (i.e., [4,9]). Considering the fact that the development and recovery of muscle fatigue heavily rely on the characteristics of the task (6,10), it is still unclear how the central and the peripheral sites of the neuromuscular system respond in the course of BFR exercise and how these sites acutely recover from such strong impairments. From a practical point of view, knowledge about the recovery process after low-intensity BFR exercise is crucial to understand the exercise–adaptation cycle to determine the optimal balance between training and recovery. Therefore, the present study was designed to provide mechanistic insight into the time course of changes in neuromuscular function during and after exercise under conditions of limited blood flow. By using various electrical stimulation methods at short time intervals during and after each exercise condition, we were able to investigate the effect of low-intensity exercise with BFR on central and peripheral aspects of muscle fatigue development and recovery.

METHODS

Subjects. Seventeen healthy males volunteered to participate in this study. A sample comprising exclusively male subjects was chosen based on the common finding that the level of muscle fatigue differs between sexes (for a review, see [11]). All subjects were physically active (training more than four times per week) and regularly engaged in a total body resistance training program for at least two times per week. Descriptive data of the subject characteristics are presented in Table 1. Participants were excluded if they were hypertensive (>140/90 mm Hg) or had more than one risk factor for thromboembolism (12). Subjects were asked to refrain from vigorous exercise, analgesics, caffeine, and alcohol consumption for 24 h before the investigations. The study was approved by the university ethics committee and was conducted according to the Declaration of Helsinki. All subjects were informed about possible risks and discomfort associated with the investigations before giving their written consent to participate.

TABLE 1. Subject characteristics.

Age (yr)	25 ± 4
Height (cm)	184 ± 6
Body mass (kg)	84 ± 6
Physical activity ($h \cdot wk^{-1}$)	9 ± 3
Resistance training ($h \cdot wk^{-1}$)	5 ± 2
SBP (mm Hg)	132 ± 9
DBP (mm Hg)	76 ± 8
Arterial occlusion pressure (mm Hg)	208 ± 25
BFR pressure (mm Hg)	125 ± 15
Unilateral 1RM (N·m)	240 ± 32

Values are expressed as means ± SD.
SBP, systolic blood pressure; DBP, diastolic blood pressure.

Experimental procedure. All subjects visited the laboratory on three different occasions. During the first visit, subjects' knee extension 1RM and arterial occlusion pressure were determined. Furthermore, subjects were thoroughly familiarized with the following procedures: (i) RPE and leg muscle pain, (ii) neuromuscular testing procedures comprising maximal voluntary contractions (MVC) and peripheral nerve stimulation, (iii) metronome pacing of knee extension exercise, and (iv) two submaximal sets (2 sets of 10 repetitions at 30% 1RM) of knee extension exercise under BFR at 60% arterial occlusion pressure. Furthermore, subjects' knee extension 1RM and arterial occlusion pressure were determined within the first session. Using a randomized, counterbalanced, within-subjects design, participants underwent two experimental conditions across two separate visits: four sets (30, 15, 15, and 15 repetitions; total exercise time, 315 s) of low-intensity knee extensions (i) with BFR and (ii) without BFR (CON). Testing sessions were separated by 7 ± 1 d and took place at the same time of the day. On the basis of the protocol previously used by Froyd et al. (13), neuromuscular tests were performed before, during (immediately after each set of knee extension exercise), and 1, 2, 4, and 8 min after each experimental condition (Fig. 1). Furthermore, RPE and leg muscle pain were assessed after each set. Electromyography (EMG) data were continuously recorded during each experimental trial.

Upon arrival at the laboratory, subjects' blood pressure and arterial occlusion pressure were determined. Before baseline measurements, subjects performed an initial warm-up on a stationary bicycle (5 min; 120 W; 90 rpm) followed by a specific warm-up on the dynamometer comprising two isometric contractions for 5 s at 50%, 70%, and 90% of maximal voluntary torque (MVT; determined during the familiarization session), respectively. Neuromuscular tests comprised supramaximal electrical stimulations of the femoral nerve during and after an isometric MVC (Fig. 1B and C). All measurements were conducted on the quadriceps muscle of the dominant leg (i.e., kicking preference). During knee extension exercise and neuromuscular testing, subjects were comfortably seated and secured on a CYBEX NORM dynamometer (Computer Sports Medicine®, Inc., Stoughton, MA). The seating position was adjusted for each subject, and settings were documented for the subsequent sessions.

Determination of arterial occlusion pressure. Arterial occlusion pressure was determined in a lying position using a handheld bidirectional Doppler probe (Hadeco Bidop ES-100V3, Kawasaki, Japan), which was placed over the posterior tibial artery. Pressure was automatically adjusted using a cuff inflator system (Heidi™; Ulrich Medical, Ulm, Germany).

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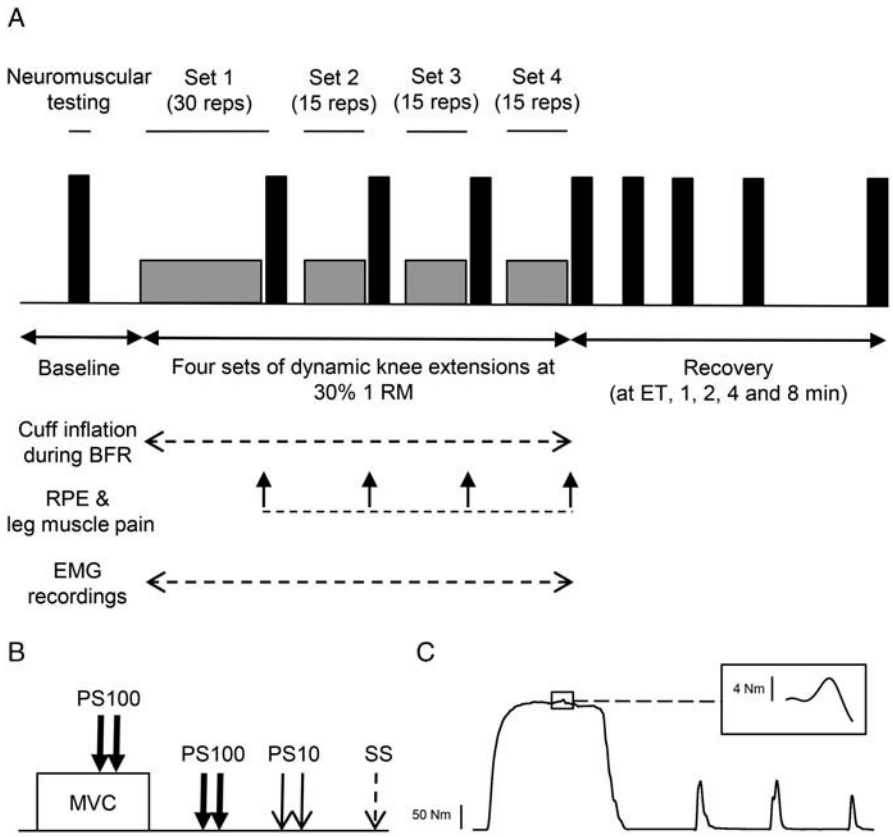

FIGURE 1—A. Illustration of the experimental design. Neuromuscular function was assessed before, during (immediately after each set of knee extension exercise), and 1, 2, 4, and 8 min after low-intensity exercise with and without BFR. Furthermore, RPE, leg muscle pain, and EMG data were recorded during each experimental condition. ET, exercise termination. B. The neuromuscular testing procedure comprised MVC of the quadriceps muscle combined with different electrical stimulation methods to assess MVT, voluntary activation (via the interpolated twitch technique), and quadriceps twitch torques in response to paired electrical stimuli at 100 Hz (PS100) and at 10 Hz (PS10) as well as single stimuli (SS). C. Typical torque recording of the neuromuscular testing procedure. An enlarged view of the interpolated twitch is presented in the box.

A 10 × 76-cm pneumatic cuff (Ulrich Medical) affixed to the most proximal part of the right thigh was incrementally inflated until the pulse of the tibial artery was interrupted. The inflation procedure was performed as described in detail by Loenneke et al. (14). Arterial occlusion pressure was determined within the familiarization session and again at baseline of each experimental condition to evaluate its reproducibility.

1RM. Subjects' unilateral knee extension 1RM was determined using the isotonic mode of a CYBEX NORM dynamometer (Computer Sports Medicine®, Inc.). 1RM was defined as the heaviest load that can be lifted through a controlled, full range of motion. Before isotonic testing, subjects performed an initial warm-up on a stationary bicycle (5 min, 120 W, 90 rpm). A second warm-up comprised six isotonic knee extensions with a submaximal load and two further contractions with a higher load. During the actual testing procedure, the load was progressively increased until 1RM was determined. Between each attempt, subjects rested for 90 s. All 1RM were determined within five attempts.

Torque recordings. A CYBEX NORM dynamometer (Computer Sports Medicine®, Inc.) was used to record instantaneous torques. Subjects were positioned on an adjustable chair with the hip fixed at 80° (0° = full extension). To avoid excessive movements during data recording, straps were fixed tightly across the subjects' waist and chest. The dynamometer rotation axis was aligned with the knee joint rotation axis, and the lever arm was attached to the lower leg just above the lateral malleolus. Isometric MVC were performed at 90° knee flexion (0° = full extension). For each trial, subjects were instructed to cross their arms in front of their chest and to push as hard and as fast as possible against the lever arm of the dynamometer. Strong verbal encouragement was given by the investigator. Visual feedback of the torque–time curve was provided on a digital oscilloscope (HM1508; HAMEG Instruments, Mainhausen, Germany).

EMG recordings. A detailed description of the EMG recordings can be found in a previously published study from our laboratory (15). Briefly, myoelectrical signals from the vastus medialis (VM), rectus femoris (RF), and vastus lateralis (VL) were recorded using surface electrodes (EMG Ambu Blue Sensor N). EMG signals were amplified (×2500), band-pass filtered (10–450 Hz), and digitized with a sampling frequency of 3 kHz using an analog-to-digital converter (NI PCI-6229; National Instruments, Austin).

Electrical nerve stimulation. To assess the neuromuscular function of the quadriceps muscle, the femoral

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nerve was stimulated percutaneously using electrical stimulation. A constant-current stimulator (Digitimer DS7A, Herfordshire, UK) was used to deliver square wave pulses of 1-ms duration with a maximal voltage of 400 V. A ball probe cathode (10-mm diameter) was pressed in the femoral triangle always by the same investigator. The anode, a self-adhesive electrode (35 × 45 mm; Spes Medica, Genova, Italy), was affixed over the greater trochanter. After determining the optimal site for stimulation, the position was marked onto the subjects' skin to ensure repeatable measurements within each session. Individual stimulation intensity was progressively increased until the maximum compound muscle action potential (M_{max}) of VM, RF, and VL as well as a plateau in knee extensor twitch torque was achieved. During the subsequent testing procedures, the stimulation intensity was increased by additional 40% to guarantee supramaximal stimulation (~50 mA). Potentiated quadriceps twitch torque evoked by paired electrical stimuli at 100 Hz (PS100) and 10 Hz (PS10) and single stimuli (SS) were elicited 2, 4, and 6 s after MVC, respectively. As recommended previously (16), quadriceps twitch torque in response to PS100 was used to characterize changes in contractile function. To determine the level of voluntary activation during isometric MVC, the interpolated twitch technique was applied (17). Electrical paired stimuli were delivered to the femoral nerve at 90° knee flexion 2 s after torque onset (during the plateau phase) and 2 s after MVC.

Exercise protocol. The exercise protocol comprised 30 repetitions of unilateral isotonic knee extensions followed by three sets of 15 repetitions at 30% of 1RM. Each set was separated by 30 s of rest. This protocol was chosen because it is typically used for research purposes and practical applications in the context of BFR (2). A metronome set at 40 bpm was used to ensure a cadence of 1.5 s for concentric and 1.5 s for eccentric muscle actions. During the BFR condition, a pneumatic cuff applied to the subjects' thigh was inflated before the first set of exercise and deflated immediately after termination of the fourth set. The target pressure was set at 60% arterial occlusion pressure.

RPE and leg muscle pain. During the first session at the laboratory, subjects were thoroughly familiarized with RPE and ratings of leg muscle pain. Subjects' perception of effort was assessed by using the 15-point Borg scale. Before each testing session, participants received written instructions based on guidelines recently proposed by Pageaux (18). Briefly, instructions included the definition of effort, exercise-specific descriptions ("How hard is it for you to drive your leg?"), exercise anchoring (i.e., "Maximal exertion corresponds to the effort you experienced while you were performing a maximal voluntary contraction"), and the distinction of effort, pain, and other exercise-related sensations (18). Leg muscle pain was recorded using a modified category-ratio 10 (CR-10) scale as proposed by Cook et al. (19). RPE and CR-10 ratings were taken immediately after each set of knee extension exercise.

Data analyses. Peak twitch torques (i.e., highest values of the torque–time curve) were determined for SS, PS10, and PS100, respectively. The PS10·PS100^{-1} torque ratio was calculated as an index of low-frequency fatigue (20). Isometric MVT was defined as the highest torque value before the superimposed twitch. M_{max} amplitudes elicited by SS were measured peak-to-peak. Muscle activity during exercise was estimated by calculating the root mean square of the EMG signal (RMS-EMG) averaged for the first three and the last three repetitions of each set, respectively (21). Only EMG data during the concentric phase of each repetition were taken into account for analysis. RMS-EMG of VM, RF, and VL were normalized to the corresponding M_{max} values (RMS·M^{-1}). The level of voluntary activation was calculated using a corrected formula: [1 – superimposed twitch (T_b × MVT^{-1}) × control twitch^{-1}] × 100 (22). MVT is the maximal torque level and T_b the torque value immediately before the superimposed twitch. The corrected formula is used to avoid the potential problem that the superimposed stimuli are not applied during the maximum torque level. Our group has recently shown that voluntary activation of the knee extensors can be reliably assessed during isometric contractions using the corrected formula (23).

Statistical analysis. All data were screened for normal distribution using the Shapiro–Wilk test. A two-way (time and condition) repeated-measures ANOVA was conducted for all neuromuscular parameters. *Post hoc* tests for all time and condition comparisons were performed with Bonferroni adjustments. The effect size was determined by calculating partial eta squared (η_p^2). Differences in RPE and leg muscle pain (CR-10) across the experimental conditions were tested using Friedman's tests. *Post hoc* analyses with Wilcoxon signed-rank tests were conducted with a Bonferroni correction applied, resulting in a significance level of $P \leq 0.0125$. Absolute and relative intersession reliability of arterial occlusion pressure was computed using an Excel spreadsheet (24). Absolute reliability was determined by computing the coefficient of variation. Relative reliability was determined by calculating the intraclass correlation coefficient (low, <0.80; moderate, 0.80–0.90; high, >0.90) (25). Data were analyzed using the SPSS statistical package 22.0 (SPSS Inc., Chicago, IL), and statistical significance was accepted at $P \leq 0.050$.

RESULTS

All participants successfully completed both exercise protocols at the required cadence.

Intersession reliability of arterial occlusion pressure. Arterial occlusion pressure of the thigh was reliably assessed as indicated by an acceptable absolute (coefficient of variation = 5.5%) and a moderate relative intersession reliability (intraclass correlation coefficient = 0.80).

MVT. A significant time × condition interaction was found for MVT ($F_{8,16}$ = 12.47, P < 0.001, η_p^2 = 0.44). Already after the first set of exercise, there was a significant group difference between BFR and CON (P = 0.048). This difference persisted up to and including the first min of recovery (all P < 0.010). A consistently greater MVT reduction was

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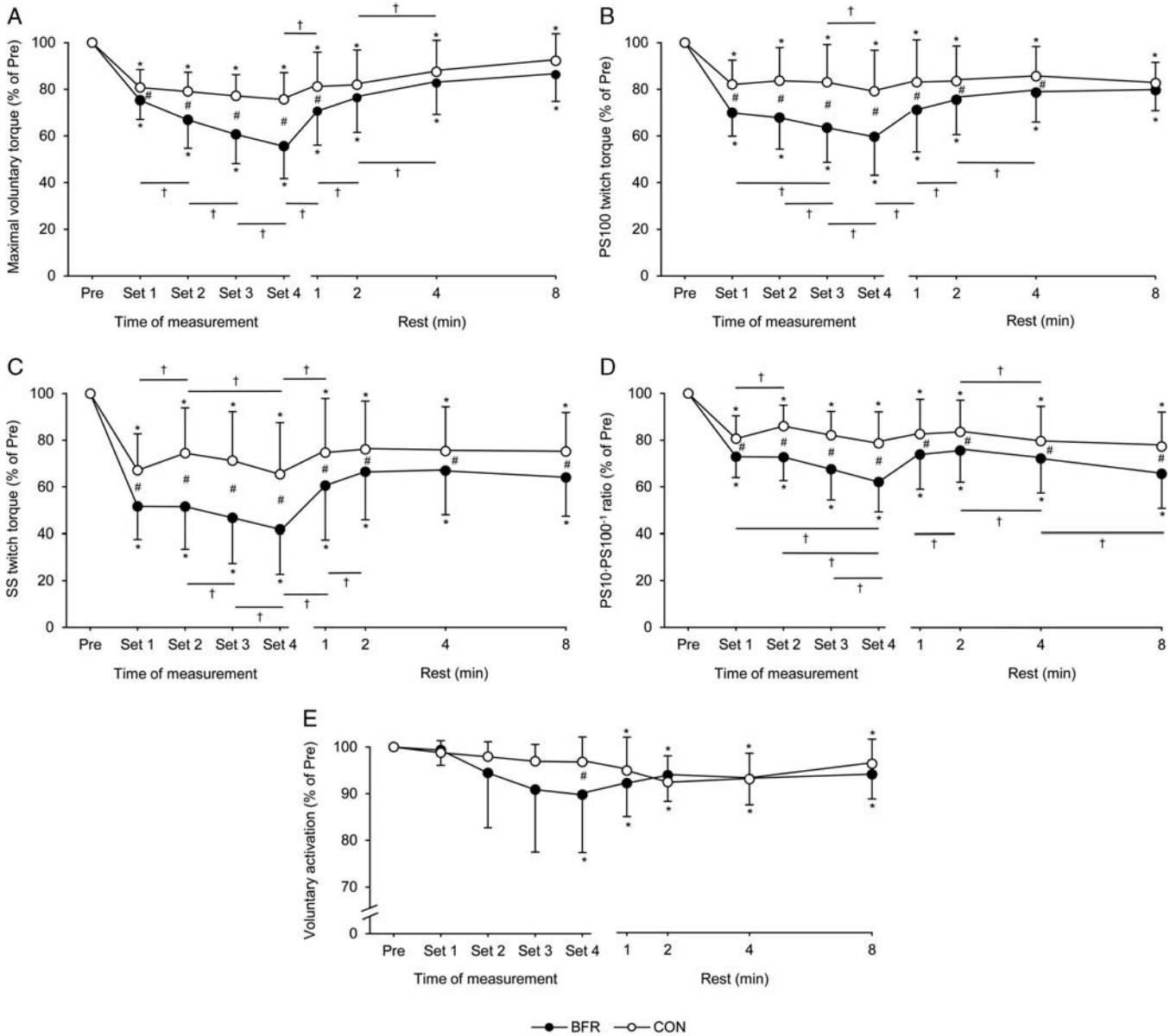

FIGURE 2—Percentage changes from baseline values for maximal voluntary torque (A), PS100 twitch torque (B), SS twitch torque (C), PS10·PS100^{-1} ratio (D), and voluntary activation (E) during and 1, 2, 4, and 8 min after each experimental condition. PS100, paired stimuli at 100 Hz; PS10, paired stimuli at 10 Hz; SS, single stimuli; PS10·PS100^{-1} ratio as an index of low-frequency fatigue; CON, control condition. Significantly different from Pre: *$P \leq 0.050$. Significantly different between time points: †$P \leq 0.050$. Significantly different between groups: #$P \leq 0.050$. Values are expressed as means ± SD. Please note that the statistics of the pairwise comparisons are presented in a reduced version to ensure clarity of the results.

observed during BFR (Fig. 2A). For both conditions, MVT differed significantly from baseline values until the eighth minute of recovery (all $P < 0.050$; Table 2). Immediately after exercise termination (set 4), there was a significantly greater decrease in MVT for BFR (−44.5% ± 14.1%) compared with CON (−24.3% ± 11.8%; $P < 0.001$). For the BFR condition, MVT progressively declined throughout the exercise protocol, recovered progressively within 4 min after exercise termination, but remained depressed by 13.4% ± 10.3% after 8 min of rest. During CON, there was a significant decrease in MVT after the first set of exercise but no further decline throughout the exercise protocol. After exercise termination, MVT values gradually recovered within 4 min postexercise but remained depressed compared with preexercise values after 8 min of rest. Absolute values and the percentage MVT changes during BFR and CON are shown in Table 2 and Figure 2A.

Electrically evoked twitch torque. There were significant time–condition interactions for PS100 ($F_{8,16}$ = 23.48, $P < 0.001$, $\eta_p^2 = 0.60$), SS ($F_{8,16}$ = 26.54, $P < 0.001$, $\eta_p^2 = 0.62$), and PS10·PS100^{-1} ratio ($F_{8,16}$ = 9.08, $P < 0.001$, $\eta_p^2 = 0.39$). After the first set of exercise, there were significant group differences for PS100, SS, and PS10·PS100^{-1} ratio between BFR and CON (all $P < 0.001$; Fig. 2B–D). Group differences for SS and PS10·PS100^{-1} ratio persisted during

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TABLE 2. Neuromuscular function of the quadriceps muscle before, during, and after each experimental condition.

			During Exercise				Rest			
		Pre	Set 1	Set 2	Set 3	Set 4	1 min	2 min	4 min	8 min
MVT (N·m)										
	BFR	292.0 ± 51.0	219.9 ± 44.5*	197.6 ± 54.3*	179.1 ± 52.4*	163.9 ± 51.4*	206.6 ± 50.5*†	222.1 ± 48.6*†	241.0 ± 48.0*†	251.4 ± 45.2*†
	CON	289.1 ± 41.6	233.3 ± 38.9*#	228.1 ± 36.0*#	222.1 ± 33.9*#	217.1 ± 36.6*#	234.3 ± 42.0*†#	236.3 ± 38.0*†	252.4 ± 38.6*†	265.7 ± 38.9†
PS100 (N·m)										
	BFR	102.5 ± 7.3	71.5 ± 10.7*	69.2 ± 13.6*	64.8 ± 14.9*	60.9 ± 16.7*	72.7 ± 18.3*†	77.3 ± 15.6*†	80.4 ± 13.5*†	81.4 ± 9.1*†
	CON	100.9 ± 11.1	82.5 ± 11.8*#	84.0 ± 14.3*#	83.0 ± 15.1*#	78.9 ± 15.2*#	83.1 ± 15.2*#	83.7 ± 12.7*#	85.9 ± 11.1*#	83.4 ± 11.5*
PS10 (N·m)										
	BFR	101.9 ± 9.8	53.1 ± 15.6*	51.9 ± 18.6*	46.1 ± 20.3*	39.9 ± 19.2*	56.3 ± 24.4*†	60.5 ± 23.2*†	59.8 ± 21.2*†	54.6 ± 18.4*†
	CON	99.2 ± 10.2	66.5 ± 17.8*#	72.5 ± 20.6*#	69.2 ± 23.2*#	63.9 ± 24.7*#	69.9 ± 25.0*†#	70.7 ± 22.1*†#	68.5 ± 19.3*#	64.0 ± 18.8*#
SS (N·m)										
	BFR	66.5 ± 5.7	34.2 ± 9.4*	34.1 ± 11.8*	30.9 ± 12.6*	27.6 ± 12.3*	39.1 ± 14.8*†	42.9 ± 13.3*†	43.3 ± 12.4*†	41.3 ± 10.9*†
	CON	65.4 ± 7.8	43.7 ± 10.7*#	48.5 ± 13.6*#	46.3 ± 13.9*#	42.7 ± 15.2*#	47.4 ± 12.9*#	48.3 ± 11.2*#	47.8 ± 9.7*#	47.6 ± 9.4*#
PS10·PS100^{-1} ratio										
	BFR	1.00 ± 0.09	0.73 ± 0.13*	0.73 ± 0.13*	0.68 ± 0.16*	0.62 ± 0.15*	0.74 ± 0.18*†	0.76 ± 0.18*†	0.73 ± 0.19*	0.66 ± 0.19*
	CON	0.99 ± 0.09	0.80 ± 0.14*#	0.85 ± 0.13*#	0.81 ± 0.15*#	0.78 ± 0.18*#	0.82 ± 0.18*#	0.83 ± 0.17*#	0.79 ± 0.18*#	0.77 ± 0.21*#
VM M_{max} (mV)										
	BFR	13.1 ± 2.8	12.7 ± 2.9	12.8 ± 2.9	12.8 ± 2.9	12.6 ± 2.9	12.7 ± 2.4	12.7 ± 2.4	12.6 ± 2.4	12.7 ± 2.5
	CON	13.1 ± 2.0	13.0 ± 2.0	13.1 ± 2.0	13.1 ± 2.0	13.2 ± 2.0	12.9 ± 1.7	13.0 ± 1.7	13.1 ± 1.8	13.0 ± 1.9
RF M_{max} (mV)										
	BFR	3.7 ± 1.1	3.8 ± 1.2	3.9 ± 1.2	3.9 ± 1.2	4.0 ± 1.2	3.9 ± 1.2	4.0 ± 1.2	3.9 ± 1.2	3.8 ± 1.2
	CON	4.1 ± 1.6	4.0 ± 1.4	4.2 ± 1.5	4.3 ± 1.4	4.3 ± 1.5	4.1 ± 1.4	4.2 ± 1.5	4.1 ± 1.5	4.0 ± 1.3
VL M_{max} (mV)										
	BFR	12.8 ± 3.8	12.7 ± 3.6	12.8 ± 3.5	12.8 ± 3.5	12.9 ± 3.6	12.3 ± 3.5	12.3 ± 3.5	11.9 ± 3.5	11.4 ± 3.5
	CON	12.9 ± 3.8	12.2 ± 3.9	12.4 ± 3.7	12.5 ± 3.6	12.6 ± 3.7	11.8 ± 3.6	11.7 ± 3.7	11.5 ± 3.8	11.2 ± 4.0
VA (%)										
	BFR	95.9 ± 3.3	95.2 ± 2.3	90.4 ± 10.7	87.1 ± 12.7	86.0 ± 11.2*	88.4 ± 7.7*	90.1 ± 6.7*	89.3 ± 6.9*	90.3 ± 6.5*
	CON	95.8 ± 3.2	94.6 ± 2.4	93.7 ± 2.2	92.8 ± 4.4	92.6 ± 4.4#	90.9 ± 4.0*	93.7 ± 2.2*	92.8 ± 4.4*	92.6 ± 4.4*

Values are expressed as mean ± SD. Note that the statistics of the pairwise comparisons are presented in a reduced version to ensure clarity of the results.
Significantly different from Pre: *$P \leq 0.050$.
Significantly different from exercise termination (set 4): †$P < 0.050$.
Significantly different between groups: #$P \leq 0.050$.
MVT, maximal voluntary torque; PS10·PS100^{-1} ratio, index of low-frequency fatigue; M_{max}, maximal M-wave; VA, voluntary activation; CON, control condition.

the entire recovery period (all $P < 0.050$; Fig. 2C–D). For PS100, group differences were still present up to and including 4 min postexercise ($P < 0.046$). At exercise termination, significantly greater reductions in PS100, SS, and PS10·PS100^{-1} ratio were found for BFR (−40.4% ± 16.9%, −58.2% ± 19.6%, and −38.0% ± 13.1%, respectively) compared with CON (−20.9% ± 17.6%, −34.6% ± 22.1%, and −21.4% ± 13.4%, respectively). For each stimulation method, twitch torques progressively declined throughout the BFR condition and recovered substantially within the first 2 min after exercise termination (Fig. 2B–D). Exercise-induced reductions in PS100 and SS persisted over the entire recovery period (all $P < 0.050$; Table 2). However, PS10·PS100^{-1} ratio progressively decreased from 2 to 8 min of recovery (Fig. 2D).

Electrically evoked potentials. No significant time–condition interactions were found for VM ($F_{8,16} = 1.32$, $P = 0.241$, $\eta_p^2 = 0.08$), RF ($F_{8,16} = 0.30$, $P = 0.964$, $\eta_p^2 = 0.02$), and VL M_{max} amplitude ($F_{8,16} = 0.9$, $P = 0.486$, $\eta_p^2 = 0.06$), respectively. Absolute values for M_{max} at each time point and condition are shown in Table 2.

Voluntary activation. A significant time–condition interaction was observed for voluntary activation ($F_{8,16} = 3.30$, $P = 0.002$, $\eta_p^2 = 0.17$). Significant group differences between BFR and CON were found after exercise termination ($P = 0.018$) with a greater reduction in voluntary activation for BFR (−10.2% ± 12.3%) compared with CON (−3.2% ± 5.5%; Fig. 2E). From the first to the eighth minute of recovery, voluntary activation for BFR and CON remained significantly reduced compared with baseline values (all $P < 0.050$; Table 2). Absolute values and percentage changes in voluntary activation during BFR and CON are presented in Table 2 and Figure 2E.

EMG recordings during exercise. There were significant time–condition interactions for VM RMS·M^{-1} ($F_{7,16} = 14.38$, $P < 0.001$, $\eta_p^2 = 0.47$) and VL RMS·M^{-1} during exercise ($F_{7,16} = 6.71$, $P < 0.001$, $\eta_p^2 = 0.30$). No significant time–condition interaction was found for RF RMS·M^{-1} ($F_{7,16} = 1.93$, $P = 0.071$, $\eta_p^2 = 0.30$). Regardless of the experimental condition, normalized muscle activity of VM and VL progressively increased during each exercise set (all $P < 0.01$; Fig. 3A and 3C). For normalized VM muscle activity, significant group differences between BFR and CON were evident for the last three repetitions of the first exercise set ($P = 0.015$); thereafter, significant differences could be documented for all time points with a consistently higher activation during BFR ($P < 0.001$; Fig. 3A). Significant group differences for normalized VL muscle activity were evident for the last repetitions of the second set ($P = 0.003$). In the following, significant differences between BFR and CON could be found for all time points with a higher activation during BFR (all $P < 0.010$; Fig. 3C).

RPE and leg muscle pain. There were significant changes in RPE over time for BFR ($\chi^2_3 = 33.28$, $P < 0.001$) and CON ($\chi^2_3 = 16.57$, $P = 0.001$). Except for the first exercise set ($P = 0.016$), RPE was significantly different between BFR and CON (all $P < 0.001$; Table 3). There were

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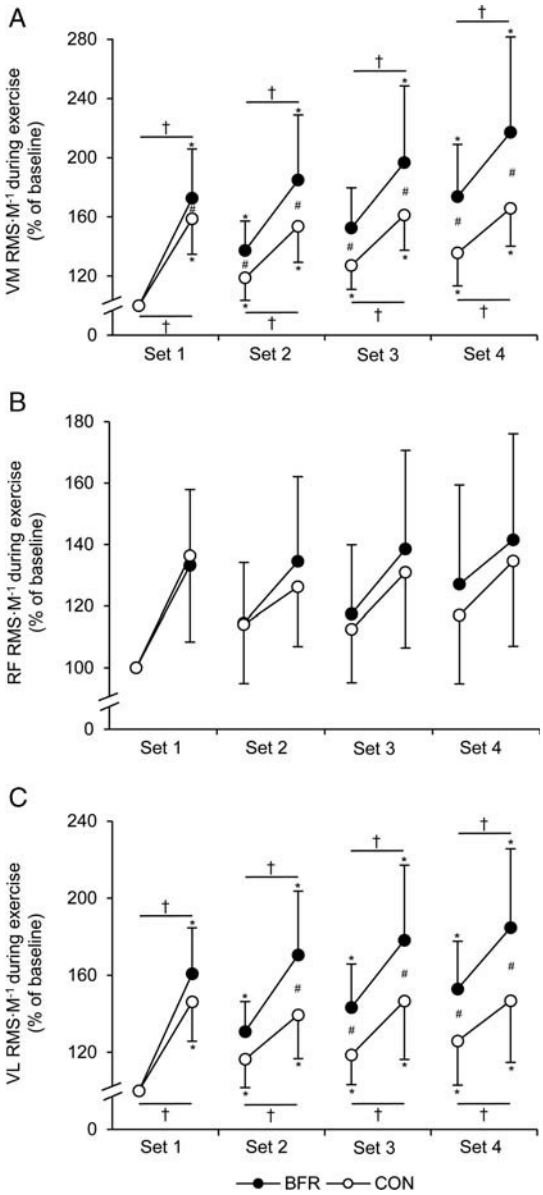

FIGURE 3—Percentage changes from baseline for the normalized muscle activity (RMS·M^{-1}) of VM (A), RF (B), and VL (C) averaged for the first three and the last three repetitions of each set. Baseline was defined as the average of the first three reps recorded during first set. RMS·M^{-1}, the root mean square of the EMG signal normalized to M_{max}; CON, control condition. Significantly different from baseline: *$P \leq 0.05$. Significantly different between time points: †$P \leq 0.050$. Significantly different between groups: #$P \leq 0.050$. Values are expressed as means ± SD.

also significant differences in leg muscle pain over time for BFR ($\chi^2_3 = 32.9$, $P < 0.001$) and CON ($\chi^2_3 = 13.90$, $P = 0.003$). Except for the first exercise set ($P = 0.040$), leg muscle pain was significantly different between BFR and CON (all $P < 0.001$; Table 3).

DISCUSSION

The present study investigated the time course and origin of changes in neuromuscular function during and after a bout of low-intensity exercise with and without BFR. The main findings were as follows: (i) BFR accelerated the exercise-induced development of muscle fatigue; (ii) peripheral factors mainly contributed to muscle fatigue during low-intensity BFR exercise with major impairments during the early phase of the exercise bout; (iii) neural factors also contributed to the pronounced end-exercise level of muscle fatigue under the condition of limited blood flow; (iv) BFR-induced muscle fatigue recovered substantially within 2 min after exercise termination; and (v) the initial recovery of muscle fatigue is mainly caused by a rapid restoration of contractile function. Interestingly, (vi) the effect of BFR on muscle fatigue was already diminished after 2 min of reperfusion. We also provide evidence that (vii) the augmented muscle activity during low-intensity BFR exercise compensated for the exacerbated contractile torque loss.

Development of muscle fatigue. As expected from other studies (i.e., [5,9]), the magnitude of muscle fatigue development during low-intensity exercise was exacerbated when blood flow to the working muscle was restricted. To our knowledge, this is the first study describing the development of muscle fatigue in the course of low-intensity exercise with and without BFR. In detail, we found a substantial loss in knee extensor MVT after the first set of exercise, irrespective of the condition. This large drop in knee extensor MVT is likely influenced by the greater number of contractions performed in the first compared with the following exercise sets (2). The exercise-induced MVT reduction was significantly greater for BFR compared with CON. In the course of the BFR condition, MVT progressively declined until exercise termination, whereas no further decrease in MVT was observed after the first set during CON. At exercise termination, there was a significantly greater MVT reduction for BFR (−45% ± 14%) compared with CON (−24% ± 12%). Studies using various exercise protocols reported BFR-induced MVT reductions of −30% to −62%, whereas minor reductions were observed for the free blood flow condition (−14% to −22% [4,9,26,27]). Discrepancies in acute responses to BFR exercise are likely due to methodological differences (i.e., cuff pressure, continuous or intermittent cuff application, exercise volume/intensity, and if exercise was performed to volitional task failure or not).

We also found that limited blood flow significantly increased the exercise-induced loss in quadriceps twitch torque, supporting previous observations that peripheral factors

TABLE 3. Perceptual responses during exercise.

		Set 1	Set 2	Set 3	Set 4
RPE					
	BFR	16 ± 2	16 ± 2	18 ± 2	18 ± 2
	CON	15 ± 2	14 ± 2*	15 ± 2*	15 ± 2*
Leg muscle pain					
	BFR	6 ± 2	6 ± 2	7 ± 2	8 ± 2
	CON	5 ± 2	4 ± 2*	5 ± 2*	5 ± 2*

Values are expressed as mean ± SD.
Significantly different between groups: *$P < 0.010$.
CON, control condition.

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mainly contribute to muscle fatigue during low-intensity exercise with BFR (9,26). In detail, there was a substantial decrease in quadriceps twitch torque after the first exercise set, irrespective of the condition. Peripheral fatigue was more pronounced during BFR compared with CON. Throughout the BFR protocol, quadriceps twitch torque progressively declined until exercise termination, whereas quadriceps twitch torque during CON remained largely unchanged after the first set. Consequently, BFR resulted in a greater end-exercise level of peripheral fatigue as indicated by a twitch torque reduction of −40% ± 17% (PS100) from the preexercise value during BFR compared with CON (−21% ± 18%). Karabulut et al. (9), who also used work-matched exercise protocols, reported a similar reduction in quadriceps twitch torque (PS100) after BFR exercise (−44%) compared with the free blood flow condition (−19%). The fact that BFR exacerbated the exercise-induced accumulation of fatigue-related metabolites and prevented the recovery of contractile function within the interset rest periods might be the most obvious explanation for the pronounced development of peripheral fatigue. To our knowledge, this is the first study which investigated the etiology of peripheral fatigue after a bout of low-intensity BFR exercise. M-wave amplitude, commonly interpreted as an index of neuromuscular propagation (28), remained unchanged in the course of the exercise protocol in both conditions. By contrast, low-frequency fatigue (estimated via $PS10 \cdot PS100^{-1}$ ratio) was more pronounced during BFR compared with CON. Together, these findings suggest that low-intensity exercise combined with BFR exacerbated the contractile torque loss likely due to alterations distal to the sarcolemma, including the direct inhibition of the cross-bridge cycle, reduced myofibrillar sensitivity to calcium (Ca^{2+}), and/or impaired Ca^{2+} release from the sarcoplasmatic reticulum (SR) (29). As shown previously (30,31), the rate of phosphocreatine hydrolysis and concomitant inorganic phosphate accumulation is accelerated when exercise is performed under conditions of limited blood flow. The latter is thought to be the main contributor to exercise-induced impairments in Ca^{2+} handling (29) and the most likely explanation for the exacerbated development of peripheral fatigue during low-intensity BFR exercise. As recently described (32), a large portion of the initial drop in quadriceps twitch torque might be explained by impairments in myofibrillar function (i.e., direct inhibition of the cross-bridge cycle and reduced Ca^{2+} sensitivity), whereas the progressive decline in muscle function during the later phases of the exercise bout might be largely explained by impaired Ca^{2+} release from the SR.

The progressive loss in quadriceps twitch torque during exercise was accompanied by significant increases in VM and VL $RMS \cdot M^{-1}$ throughout both exercise protocols. Quadriceps muscle activity, as already observed by others (i.e., [21]), was significantly augmented in the course of exercise under conditions of limited blood flow. Higher muscle activation during BFR is commonly interpreted as an increased recruitment of type II muscle fibers, which is thought to be a potential rational for hypertrophic adaptations (3). Few researchers speculated that an increased inhibitory feedback of metabosensitive muscle afferents to the alpha motoneurons resulted in an augmented fiber recruitment to maintain adequate force output (21,33). However, the most plausible explanation is that the increased muscle activation during BFR compensates for the pronounced contractile force loss during BFR exercise (34). We also found that the augmented muscle activity was accompanied by higher RPE when exercise was performed under conditions of limited blood flow. Increased effort perception as a result of low-intensity work-matched exercise with BFR was frequently reported in the literature (i.e., [5]). It is well accepted that neuronal processing of sensory signals are involved in effort perception (35). In the present study, the stronger perception of effort during BFR exercise might result from an increased central motor command and a concomitant corollary discharge to compensate for the augmented contractile dysfunction (36) and/or from an increased afferent feedback from the working muscles due to stronger metabolic disturbances in the periphery (37,38). However, the exact sensory signals generating perception of effort are still debated (36).

Furthermore, by using the interpolated twitch technique, we found that central factors also contribute to the pronounced end-exercise level of quadriceps muscle fatigue during low-intensity BFR exercise. In contrast to the strong impairments in quadriceps twitch torque already observed after the first exercise set, the reduction in voluntary activation was not evident until the last set of BFR exercise (−10% ± 12%). After exercise termination, no significant changes in voluntary activation could be found during CON. This observation is in line with Karabulut et al. (9), who reported a 13% decline in voluntary activation at exercise termination induced by five sets of 20 dynamic knee extensions at 20% 1RM with BFR and a 4% increase after CON. Together, restricted blood flow appears to promote the development of central fatigue after multiple sets of work-matched low-intensity exercise. Although several mechanisms have been proposed to cause a reduction in voluntary activation (6), one mechanism that presumably accounts for decreased motoneuron firing rates under conditions of limited blood flow is the inhibitory feedback of group III/IV muscle afferents (39). A distinction is made between ergoreceptive group III/IV muscle afferents, which respond to low levels of interstitial ATP, H^+, and lactate associated with freely perfused, mostly aerobic exercise and nociceptive muscle afferents that are sensitive to high levels of metabolites associated with painful and/or ischemic exercise (40). The fact that higher ratings of leg muscle pain were recorded during BFR compared with CON might be a plausible but indirect indicator that nociceptive group III/IV muscle afferents were activated to a greater extent when exercise is performed under conditions of limited blood flow. Furthermore, it has been shown that group IV afferents can also be stimulated by venous distension (41); hence, it is conceivable that BFR-induced venous pooling itself has contributed to increased

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discharge rates. Because group III/IV afferents are thought to decrease neural drive by acting at the spinal and/or supraspinal level, changes at spinal and/or supraspinal sites might have contributed to central fatigue after multiple sets of BFR exercise. However, by using peripheral nerve stimulation to assess voluntary activation, we were unable to determine whether spinal and/or supraspinal factors contributed to impairments in neural drive immediately after BFR exercise.

Recovery of muscle fatigue. To our knowledge, this is the first study investigating the time course and origin of neuromuscular recovery after fatiguing low-intensity exercise with and without BFR. We found a progressive but incomplete recovery of maximal voluntary quadriceps strength within 8 min postexercise, irrespective of the condition. Compared with CON, reperfusion after BFR exercise induced a markedly faster restitution of maximal voluntary quadriceps strength within the first 2 min after exercise termination. Despite the exacerbated muscle fatigue during BFR exercise, group differences were no longer evident after 2 min of rest, suggesting that the effect of limited blood flow on muscle fatigue was abolished shortly after reperfusion. However, MVT values did not recover completely within 8 min of rest. This observation is in accordance with Loenneke et al. (5), who measured maximal voluntary quadriceps strength 1, 24, and 48 h after the same BFR exercise protocol and found that MVT reductions persisted for 1 h postexercise and were no longer significantly different after 24 h.

The initial recovery of muscle fatigue after BFR exercise was mainly determined by peripheral factors as indicated by a rapid restitution of quadriceps twitch torque within the first 2 min after blood flow was restored. Despite the smaller extent to which muscle fatigue developed during CON, quadriceps twitch torque did not recover significantly after exercise termination. Interestingly, the effect of limited blood flow on contractile function was no longer evident after 8 min of reperfusion. The initial restitution of contractile function with reperfusion can be largely explained by the recovery of metabolically induced impairments in intracellular Ca^{2+} handling and/or sensitivity as indicated by a significant rebound of the $PS10{\cdot}PS100^{-1}$ ratio during the first 2 min after exercise termination. The fast initial restitution of quadriceps twitch torque might also be facilitated by a reactive hyperemic blood flow after cuff deflation (42), which, in turn, could have accelerated the removal of fatigue-related metabolites. However, contractile function did not recover completely within 8 min postexercise. This is not unusual because incomplete muscle function was shown to persist for some hours due to prolonged impairments in intracellular Ca^{2+} release from the SR or myofibrillar Ca^{2+} sensitivity (29). The exercise-induced production of reactive oxygen/nitrogen species has been recently linked to those prolonged impairments in contractile function (43).

It is important to emphasize that there was a progressive decline in $PS10{\cdot}PS100^{-1}$ ratio after 2 min of rest, whereas quadriceps twitch torque in response to SS and PS100 slightly increased or remained unchanged. This is in line with data from Froyd et al. (13), who also observed a decline in $PS10{\cdot}PS100^{-1}$ ratio after 2 min of recovery. This rather contradictory behavior might question the validity of the $PS10{\cdot}PS100^{-1}$ ratio as an index of low-frequency fatigue during the later phases of the recovery period after single-joint exercise.

Compared with the rapid restitution of quadriceps twitch torque shortly after reperfusion, there was no significant recovery of voluntary activation after BFR exercise. Group differences between BFR and CON were only evident at exercise termination and disappeared 1 min postexercise. Therefore, the present data suggest that voluntary drive is only affected when multiple sets of low-intensity exercise were performed under conditions of restricted blood flow and that this effect disappeared shortly after reperfusion. In line with the present findings, studies using postexercise ischemia to investigate the effect of group III/IV muscle afferents on central factors of muscle fatigue have demonstrated that the restoration of blood flow rapidly abolished the inhibitory effects of metabosensitive muscle afferents on voluntary activation (44,45). However, voluntary activation was significantly reduced 1 min postexercise, irrespective of the condition, suggesting that multiple sets of low-intensity dynamic exercise *per se* impaired neural drive to the quadriceps muscle, which, in turn, is not evident immediately after exercise termination. Reductions in neural drive persisted for the entire recovery period, irrespective of the condition. This is in line with studies showing a slow and incomplete recovery of voluntary activation after sustained low-intensity isometric contractions (for a review, see Carroll et al. [10]). Furthermore, because voluntary drive has been found to recover slowly after eccentric muscle actions (46), the eccentric portions might have contributed to the long-lasting depression of neural drive observed in the present study. The mechanisms underlying this delayed recovery of central fatigue after low-intensity contractions are currently unknown (10).

Limitations. A limitation to highlight is that the present findings are limited to male subjects. Previous research by Labarbera et al. (47) suggested that females are less fatigable compared with males, even when performing isotonic knee extensions under conditions of limited blood flow (47). Further studies are therefore needed to understand the role of sex during muscle fatigue in the time course of low-intensity BFR exercise.

CONCLUSION

The present study provides, for the first time, mechanistic insight into the etiology of muscle fatigue development and recovery when low-intensity exercise is performed under conditions of limited blood flow. We found that BFR accelerated the development of muscle fatigue mainly due to pronounced impairments in contractile function. The major change in contractile function occurred early during the BFR exercise bout, whereas the impairment in neural drive did

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not play a significant role until exercise termination. Despite the pronounced level of muscle fatigue during BFR exercise, the effect of limited blood flow on muscle fatigue was diminished after 2 min of reperfusion, suggesting that BFR has a strong but short-lasting effect on neuromuscular function of the quadriceps muscle. The strong decline in neuromuscular function and the fast recovery after low-intensity BFR exercise seem to provide a strong adaptive stimulus for muscular growth without long-lasting impairments in motor performance, which are typically associated with heavy-load resistance training. From a practical point of view, low-intensity BFR exercise should be favored when applying high-frequency training regimes for muscle hypertrophy.

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